Vinayaka A Belludi

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Computer Vision - ML/DL

EDUCATION

KLE Technological University **2020 – 2025**
Bachelor of Engineering *Hubli, Karnataka | CGPA - 7.5*

TECHNICAL SKILLS

Languages: Python, C/C++

Frameworks: PyTorch, OpenCV, Pandas, Numpy, O3D, Scikit-Learn, Rasterio, Gdal

Platforms: Linux, VS Code, Jupyter, Unity, SNAP, Qgis

Expertise: AI/ML-DL, Computer Vision, 3D Reconstruction, Image Processing, Data Science, GIS, Model Post Processing, Game Development

EXPERIENCE

Arka Creatos **June 2023 – November 2023**
Computer Vision Intern *Hubli, Karnataka*
- Developed an Oral Cancer Segmentation Model, achieving a mean Intersection over Union (mIoU) score of 0.47.
- Implemented automated reporting systems for doctors, reducing report generation time by 70%.
- Designed and developed a user-friendly application interface using PyQt5 for real-time photo and video capture with integrated cancer segmentation and automated report generation.

Dfy-graviti **November 2024 – Present**
AI/ML Intern (with Indian Navy) *Banglore, Karnataka*
- Developed detection model achieving 83% precision and 72% recall for maritime surveillance using multispectral imagery.
- Enhanced vessel identification, raising small/medium ship detection rates from 20% to 80%.
- Designed end-to-end system workflow integrating data processing, model inference and reporting.

PROJECTS

Neural Implicit 3D Reconstruction for Indoor Scenes | PyTorch, NeRFs, 3D Reconstruction **June 2024**
- Implemented the paper MonoSDF to achieve high-quality 3D reconstructions from 2D images.
- Employed depth and normal maps to refine the spatial understanding of 3D scenes, enhancing accuracy by 15%.
- Generated SDFs from volumetrically rendered ray distances for 3D object boundaries.
- Currently working on novel enhancements to further improve impact and accuracy.

Denoising 3D Point Cloud using Implicit Functions | PyTorch, Point Clouds, Noise Removal **February 2024**
- Implemented the paper Points2Surf for denoising 3D point clouds using implicit functions.
- Used SDF estimation to remove noise and refine surface representations, significantly improving clarity.
- Applied the Marching Cube algorithm for efficient point sampling and accurate 3D surface representation.

Semantic Segmentation on IDD | TensorFlow, Semantic Segmentation, Image Processing **May 2023**
- Employed deep learning models for semantic segmentation on the Indian Driving Dataset.
- Trained a UNet model for 150 epochs, achieving a training accuracy of 38%.
- Trained a PSPNet model for 50 epochs, achieving a training accuracy of 78%.

RESEARCH PUBLICATIONS

"AfforDrive: Detection of Drivable Area for Autonomous Vehicles" **December 2023**
- Presented at PReMI 2023 and available in Springer. Introduces a novel Visual Affordance approach for detecting drivable areas, advancing autonomous navigation and safety.

"VisionCraft: Advanced Image Captioning with Pre-trained Models" **June 2024**
- Accepted at ICMISC 2024 and to be published in Springer. Presents a multi-modal approach leveraging pre-trained models for improved image caption generation.

Independent Research **Ongoing**
- Independent researcher collaborating with Qualcomm and other companies CV researchers from USA
- Focusing mainly on computational photography, depth refinement.

CERTIFICATIONS & ACHIEVEMENTS

Game Development Mini Degree **Jul 2022**
- Earned 25 ECTS at XAMK Open University, Finland; developed 3 games.

CVPR Workshop Participation **Jun 2023**
- Contributed to a semantic segmentation project on a 30K-image dataset (27 classes).